

1 · Supplement: sets, measurable maps and Bertrand's paradox

Data Analytics for AI · Executive M.Tech · IIT Jammu

Main lecture notes · Sampling-map demonstration

This supplement gives the formal definitions, parametrizations and derivations supporting the main lecture. Its details are intended for supplementary study alongside the corresponding lecture topics. The notation agrees with the lecture notes: R is the fixed circle radius, T the normalized midpoint distance, and X the chord length.

Reference: Krishna Jagannathan, *Probability Foundations for Electrical Engineers*, 9 July 2025, Chapters 1, 4 and 11, particularly Definitions 68–70 for measurable functions, random variables and their laws. The calculations of the sampling maps are written out below.

1. Outcomes, events and mutual exclusivity

A sample space is required to satisfy two conditions. It must be **exhaustive**, so that at least one of its elements occurs in any realization of the experiment, and its elements must be **mutually exclusive**, so that at most one of them occurs. Together these state that exactly one $\omega \in \Omega$ occurs each time the experiment is performed.

Mutual exclusivity of outcomes is a property of the singleton events:

$$\{\omega_1\} \cap \{\omega_2\} = \emptyset \quad \text{whenever } \omega_1 \neq \omega_2.$$

For countable Ω the singletons therefore form a countable disjoint partition, $\Omega = \bigcup_{\omega \in \Omega} \{\omega\}$.

Outcomes are mutually exclusive; events need not be. For a single roll of a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$, and the outcomes 2 and 4 cannot both occur. The events

$$A = \{2, 4, 6\} \text{ (even)}, \quad B = \{4, 5, 6\} \text{ (greater than 3)}$$

can both occur, since $A \cap B = \{4, 6\} \neq \emptyset$.

Consequences for calculation

The two cases are handled by different rules, and the distinction above is what separates them.

Because singletons are disjoint, countable additivity applies directly to them, and for countable Ω

$$P(A) = \sum_{\omega \in A} P(\{\omega\}).$$

For two events no such rule is available, since additivity requires disjointness. The correction term is the shared part:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Writing $P(A \cup B) = P(A) + P(B)$ for the die events above would count the outcomes 4 and 6 twice.

Two errors that this distinction prevents

A list of events is not a sample space. For one roll of a die, the pair "even" and "greater than 3" is not a sample space: both can occur in a single roll, so the second condition fails. They are events in the finer space $\{1, \dots, 6\}$.

A valid sample space need not have equally likely elements. In the two-toss example of Section 5.1 of the main notes, the list "a head at once", "a tail then a head", "two tails" is exhaustive and mutually exclusive, so it is a legitimate sample space. Its members carry probabilities $1/2$, $1/4$ and $1/4$. Mutual exclusivity is a requirement on the sample space; it says nothing about the probabilities assigned to its elements, and counting the members is valid only when they happen to be equally probable.

The same reading in the Bertrand construction

Each performance of any of the three sampling procedures produces exactly one chord. The chords are therefore the elementary outcomes, they are mutually exclusive, and $\mathcal{C}_*$ is the sample space. The three models agree on this sample space and differ only in the measure placed on it.

The event $E = \{c: X(c) > \sqrt{3} R\}$ is not an outcome. It is a set containing uncountably many chords, and it is assigned a probability as a set.

This last point also marks the limit of the summation rule above. In a continuous model each individual outcome has probability zero, so

$$\sum_{c \in E} P(\{c\}) = 0 \neq P(E),$$

and the probability of E cannot be recovered by adding the probabilities of the outcomes it contains. Countable additivity is additivity over countably many disjoint events, and E is not a countable union of its singletons. The rule $P(A) = \sum_{\omega \in A} P(\{\omega\})$ is confined to countable sample spaces for this reason.

2. Random variables, inverse images and probability laws

On a probability space $(\Omega, \mathcal{F}, P)$, a real-valued **random variable** is a measurable function

$$X: \Omega \rightarrow \mathbb{R}.$$

Measurability means that $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \in \mathcal{B}(\mathbb{R})$. The Borel σ -algebra is generated by open intervals and includes the intervals, singleton sets and their countable combinations used here. For a real-valued function, it is sufficient to check $\{\omega: X(\omega) \leq x\} \in \mathcal{F}$ for every real x .

The distinction between a random variable and a set function

A random variable is a function **on outcomes**, not a real-valued set function on events. It induces the following map between collections of sets:

$$X^{-1}: \mathcal{B}(\mathbb{R}) \rightarrow \mathcal{F}, \quad B \mapsto \{\omega: X(\omega) \in B\}.$$

The probability measure and the distribution of X are real-valued **set functions**:

$$P: \mathcal{F} \rightarrow [0, 1], \quad P_X: \mathcal{B}(\mathbb{R}) \rightarrow [0, 1],$$

$$P_X(B) = P(X^{-1}(B)) = P(X \in B).$$

Thus a numerical question about X is first converted to a set of outcomes, and that event is assigned a probability. This is the set-based construction used in the Bertrand calculation below. It follows the distinction in Jagannathan, Chapter 11, Definitions 68–70.

Worked example: the head count

For two independent fair tosses, each outcome in $\Omega = \{HH, HT, TH, TT\}$ has probability $1/4$, and $\mathcal{F} = 2^\Omega$. Define $X(HH) = 2$, $X(HT) = X(TH) = 1$, and $X(TT) = 0$. Then

$$P_X(\{1\}) = P(\{HT, TH\}) = 1/2,$$

$$P_X([1, 2]) = P(\{HH, HT, TH\}) = 3/4.$$

The distribution assigns probabilities $1/4, 1/2, 1/4$ to $0, 1, 2$. It is obtained through inverse images, not by assuming that the three numerical values are equally likely.

Worked example: an indicator

For an event $A \in \mathcal{F}$, its indicator is the random variable

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Its inverse images satisfy $\mathbf{1}_A^{-1}(\{1\}) = A$ and $\mathbf{1}_A^{-1}(\{0\}) = A^c$. Consequently, $P(\mathbf{1}_A = 1) = P(A)$.

In the parity model of Section 4 of the main notes, the indicator of an even face is measurable. The numerical face-value function is not measurable relative to that σ -algebra, since the inverse image of $\{6\}$ is the unavailable subset $\{6\}$.

3. Bertrand's paradox: sampling maps and induced distributions

This section is the whole of the Bertrand material for Lecture 1. It is supplementary: the main notes use it only as an illustration in Section 5 and do not develop it. The summary below specifies the three models and states their answers; the subsections after it derive those answers, together with the parametrizations, inverse formulas and densities.

Summary: the three models and their answers

Chords, coordinates and the event

Fix a circle of radius R . For a chord c , let $m(c)$ be its midpoint. Use the coordinates

Here Arg is taken in $[0, 2\pi)$.

$$\phi = \text{Arg}(m(c)), \quad t = \|m(c)\|/R.$$

For $0 < t < 1$, the forward map $C(\phi, t)$ places the midpoint at $Rt(\cos\phi, \sin\phi)$ and draws the chord perpendicular to that radius. Its inverse recovers the midpoint angle and normalized distance:

$$C^{-1}(c) = (\text{Arg}(m(c)), \|m(c)\|/R).$$

Take $0 \leq \phi < 2\pi$. Diameters and zero-length chords are excluded from this chart; they have probability zero under all three sampling rules. Explicit endpoint formulas and the measurable spaces are given below.

The **chord-length random variable** is

$$X(c) = 2R\sqrt{1 - T(c)^2}, \quad T(c) = \|m(c)\|/R.$$

The side of the inscribed equilateral triangle has length $\sqrt{3} R$. Thus the event being measured is

$$E = \{c: X(c) > \sqrt{3} R\} = \{c: T(c) < 1/2\}.$$

Sampling distributions, maps and inverses

Let U, V be independent uniform random variables on $(0, 1)$ and $[0, 1)$, respectively. Write Q for their joint uniform probability measure. The three sampling maps have the form

$$\Psi_j(u, v) = C(2\pi v, q_j(u)).$$

Model	Sampling rule	$t = q_j(u)$	Inverse $u = q_j^{-1}(t)$
A	Independent uniform endpoints	$\cos(\pi u/2)$	$(2/\pi)\arccost$
B	Uniform radial distance	u	t
C	Uniform midpoint in disk area	$\sqrt{u}$	t^2

In every row, the angular inverse is $v = \phi/(2\pi)$. Consequently, a sampled chord determines its sampling coordinates through

$$\Psi_j^{-1}(c) = \left(q_j^{-1}(T(c)), \frac{\text{Arg}(m(c))}{2\pi} \right).$$

For A, the original endpoint angles are independent and uniform on $[0, 2\pi)$. The half-angle H of the minor arc is uniform on $(0, \pi/2)$, so $U = 2H/\pi$ is uniform and $T = \cos H$. The midpoint direction is independent and uniform. The original map from **ordered** endpoints to an unordered chord is two-to-one. The table uses the equivalent one-to-one midpoint/half-angle parametrization. Its justification and the recovery of the unordered endpoints follow below.

For B, uniform radial distance gives T a uniform distribution on $(0, 1)$; it can be generated by setting $T = U$.

For C, uniform disk area implies $P(T \leq t) = \pi(Rt)^2/(\pi R^2) = t^2$, so T has the same distribution as $\sqrt{U}$. This can be generated by setting $T = \sqrt{U}$. Setting $T = U$ would instead specify Model B.

Inverse images give the three probabilities

The induced chord measure is $P_j(F) = Q(\Psi_j^{-1}(F))$ for every measurable chord event F . The length random variable on the uniform source space is $X_j = X \circ \Psi_j$.

Model	Length $X_j(u, v)$	Event $\Psi_j^{-1}(E)$	Probability
A	$2R\sin(\pi u/2)$	$u > 2/3$	1/3
B	$2R\sqrt{1-u^2}$	$u < 1/2$	1/2
C	$2R\sqrt{1-u}$	$u < 1/4$	1/4

For example, in A,

$$X_A > \sqrt{3} R \Leftrightarrow \sin(\pi U/2) > \sqrt{3}/2 \Leftrightarrow U > 2/3.$$

Since U is uniform, this event has probability $1 - 2/3 = 1/3$. The other two rows follow by solving their inequalities in U .

The same chord event E has different probabilities because P_A, P_B, P_C are different measures. On the common uniform source space, the event preimages $\Psi_j^{-1}(E) = \{(u, v): X_j(u, v) > \sqrt{3} R\}$ are different sets, with Q -measures 1/3, 1/2, and 1/4. A coordinate change within one fixed model preserves event probabilities.

A coordinate change within a fixed model

Keep Model B and put $W = T^2$. Its inverse is $T = \sqrt{W}$. Then

$$P_B(W < 1/4) = P_B(T < 1/2) = 1/2.$$

In this model, $P_B(W \leq w) = \sqrt{w}$, so W is not uniform. Declaring W uniform would change the sampling distribution to Model C. Correct reparametrization preserves probability; changing the distribution need not.

Interactive sampling maps and inverse images

Derivations

The remainder of this section establishes the statements above.

The event and a common chord space

Fix a circle of radius $R > 0$. The side of an inscribed equilateral triangle has length $\sqrt{3} R$. For a chord c , define the real-valued random variables

$$D(c) = \text{distance from the centre to } c, \quad T(c) = D(c)/R, \quad X(c) = \text{length of } c.$$

The geometry gives

$$X(c) = 2R\sqrt{1 - T(c)^2}, \quad E = \{c: X(c) > \sqrt{3} R\} = \{c: T(c) < 1/2\}.$$

We compare the probability of this **same event** under three probability measures on a common chord space.

For a one-to-one parametrization, let $\mathcal{C}_*$ be the set of unordered, nondegenerate chords that are not diameters. Coincident-endpoint chords and diameters have probability zero under all three rules below. Excluding them makes the inverse formulas unambiguous without changing any calculated probability.

They may instead be retained in an enlarged space with total probability zero.

A chord parametrization and its inverse

Let $0 < t < 1$ and $0 \leq \phi < 2\pi$. Write

$$n(\phi) = (\cos\phi, \sin\phi), \quad n_{\perp}(\phi) = (-\sin\phi, \cos\phi).$$

Define $C(\phi, t)$ to be the chord with unordered endpoints

$$z_{\pm} = Rt n(\phi) \pm R\sqrt{1-t^2} n_{\perp}(\phi).$$

Its midpoint is $m = Rt n(\phi)$, so the inverse is

$$C^{-1}(c) = \left(\text{Arg}(m), \frac{\|m\|}{R} \right), \quad m = \frac{z_+ + z_-}{2}.$$

Here Arg is chosen in $[0, 2\pi)$. The midpoint is nonzero because diameters were excluded. Hence $C: [0, 2\pi) \times (0, 1) \rightarrow \mathcal{C}_*$ is a bijection. Give $\mathcal{C}_*$ the σ -algebra transported from the Borel sets of this parameter domain; this makes C and its inverse measurable.

The length function X is **not** a bijection on chords: all orientations with the same t have the same length. The inverse image $X^{-1}(B)$ is still defined for every Borel set of lengths B .

A common source of uniform samples

Use the probability space

$$\Omega_0 = (0, 1) \times [0, 1), \quad \mathcal{F}_0 = \mathcal{B}(\Omega_0), \quad Q = \text{uniform area measure on } \Omega_0.$$

The coordinate random variables $U(u, v) = u$ and $V(u, v) = v$ are independent uniform variables. Endpoints of these intervals have measure zero. For each mechanism $j \in \{A, B, C\}$, define a sampling map

$$\Psi_j(u, v) = C(2\pi v, q_j(u)).$$

The induced probability measure on chord events is

$$P_j(F) = Q(\Psi_j^{-1}(F)), \quad F \in \mathcal{F}_{\mathcal{C}_*}.$$

The chord-length random variable on the uniform source space is $X_j = X \circ \Psi_j$. We now specify each q_j , its inverse and the distribution it produces.

A. Two independent uniform endpoints

The original procedure selects independent angles $A_1, A_2 \sim \text{Unif}[0, 2\pi)$ and joins the points $R(\cos A_i, \sin A_i)$. The joint angle density is $1/(2\pi)^2$.

Let $\delta \in (-\pi, \pi]$ be the signed principal difference $A_2 - A_1$ modulo 2π , and let $H = |\delta|/2$. The endpoint case $|\delta| = \pi$ has probability zero. The map $(a_1, a_2) \mapsto (a_1, \delta)$ is a bijection apart from this null boundary and the angular wrap. For each a_1 , the second coordinate is a translation modulo 2π , so the map preserves area. Hence (A_1, δ) has constant density $1/(4\pi^2)$ on the rectangle.

Now define $\Phi = (A_1 + \delta/2) \bmod 2\pi$. The map $(a_1, \delta) \mapsto (\Phi, \delta)$ is also one-to-one on the chosen half-open angular ranges and preserves area: for fixed δ it is a translation in the first coordinate. Thus (Φ, δ) has the same constant density, which factors into uniform marginal densities. Applying $H = |\delta|/2$ gives the joint density

$$f_{\Phi, H}(\phi, h) = \frac{1}{2\pi} \frac{2}{\pi}, \quad 0 \leq \phi < 2\pi, \quad 0 < h < \pi/2.$$

Therefore Φ and H are independent, Φ is uniform, and $H \sim \text{Unif}(0, \pi/2)$. The midpoint is in direction Φ , since the midpoint vector is proportional to $\cos(H) n(\Phi)$ and $\cos(H) > 0$ almost surely.

The ordered-endpoint map is two-to-one because interchanging endpoints leaves the chord unchanged. Its inverse therefore recovers an **unordered pair**, not a unique ordered pair. From (ϕ, t) , put $h = \arccost$; the unordered endpoint angles are

$$\{(\phi - h) \bmod 2\pi, (\phi + h) \bmod 2\pi\}.$$

Thus an equivalent, bijective sampling parametrization of the unordered chords uses $U = 2H/\pi$ and $V = \Phi/(2\pi)$:

$$q_A(u) = \cos(\pi u/2), \quad \Psi_A^{-1}(c) = \left(\frac{2}{\pi} \arccos T(c), \frac{\text{Arg}(m)}{2\pi} \right).$$

The length random variable and the inverse image of the event are

$$X_A(u, v) = 2R \sin(\pi u/2), \quad \Psi_A^{-1}(E) = \{(u, v): u > 2/3\}.$$

Consequently,

$$P_A(E) = Q(U > 2/3) = 1/3.$$

For $0 < t < 1$, the induced distribution of normalized midpoint distance is

$$F_{T,A}(t) = 1 - \frac{2}{\pi} \arccost = \frac{2}{\pi} \arcsint, \quad f_{T,A}(t) = \frac{2}{\pi \sqrt{1-t^2}}.$$

It is not uniform in radial distance.

B. Uniform radial distance

Choose $D \sim \text{Unif}(0, R)$ and an independent direction $\Phi \sim \text{Unif}[0, 2\pi)$. Construct the perpendicular chord through $D n(\Phi)$. Equivalently,

$$q_B(u) = u, \quad \Psi_B^{-1}(c) = \left(T(c), \frac{\text{Arg}(m)}{2\pi} \right).$$

The original (D, Φ) joint density is $1/(2\pi R)$ on $(0, R) \times [0, 2\pi)$. In normalized coordinates, T has density $f_{T,B}(t) = 1$ on $(0, 1)$.

$$X_B(u, v) = 2R\sqrt{1-u^2}, \quad \Psi_B^{-1}(E) = \{(u, v): u < 1/2\}.$$

Therefore

$$P_B(E) = Q(U < 1/2) = 1/2.$$

C. Uniform midpoint in disk area

Choose a midpoint $M = (M_1, M_2)$ with Cartesian density $1/(\pi R^2)$ inside the disk. In polar coordinates $M = Rt n(\phi)$, the area element is $R^2 t dt d\phi$. Hence

$$f_{T,\Phi}(t, \phi) = \frac{t}{\pi} = 2t \cdot \frac{1}{2\pi}, \quad 0 < t < 1.$$

Thus T and Φ are independent, Φ is uniform, and T has density $2t$ and distribution function t^2 . A uniform draw U is transformed by

$$q_C(u) = \sqrt{u}, \quad \Psi_C^{-1}(c) = \left(T(c)^2, \frac{\text{Arg}(m)}{2\pi} \right).$$

The length random variable and event preimage are

$$X_C(u, v) = 2R\sqrt{1-u}, \quad \Psi_C^{-1}(E) = \{(u, v): u < 1/4\}.$$

Therefore

$$P_C(E) = Q(U < 1/4) = 1/4.$$

The area ratio $\pi(R/2)^2/(\pi R^2)$ gives the same result.

Explicit comparison on the same coordinates

In every model, Φ is independent of T and uniform on $[0, 2\pi)$. The density with respect to $d\phi dt$ is $f_{T,j}(t)/(2\pi)$.

Model	$T = q_j(U)$	Inverse $U = q_j^{-1}(T)$	Density $f_{T,j}(t)$, $0 < t < 1$	$P_j(E)$
Uniform endpoints	$\cos(\pi U/2)$	$(2/\pi)\arccos T$	$2/(\pi\sqrt{1-t^2})$	1/3
Uniform radial distance	U	T	1	1/2
Uniform midpoint area	$\sqrt{U}$	T^2	$2t$	1/4

For each row,

$$P_j(E) = \int_0^{1/2} f_{T,j}(t) dt = Q(\Psi_j^{-1}(E)).$$

There is no equality between $P_A(E)$, $P_B(E)$ and $P_C(E)$ to be contradicted: the measures are different. On the common uniform source space, the event preimages $\Psi_j^{-1}(E) = \{(u, v): X_j(u, v) > \sqrt{3} R\}$ are different sets, with Q -measures 1/3, 1/2, and 1/4. Within a fixed model, a correct reparametrization preserves event probability.

A reparametrization within one fixed model

Retain Model B and set $W = T^2$. Since T is uniform on $(0, 1)$,

$$P_B(W \leq w) = P_B(T \leq \sqrt{w}) = \sqrt{w}, \quad f_{W,B}(w) = \frac{1}{2\sqrt{w}}, \quad 0 < w < 1.$$

The inverse transformation is $T = \sqrt{W}$. The same event is now $W < 1/4$, and

$$P_B(E) = \int_0^{1/4} \frac{dw}{2\sqrt{w}} = \frac{1}{2}.$$

If W were instead declared uniform, its density would be changed to 1. That would give Model C and probability 1/4. This would be a change of probability model, not merely a change of coordinates.

The distribution of chord length (additional calculation)

For $0 \leq x \leq 2R$, put $z = x/(2R)$. Because $X = 2R\sqrt{1 - T^2}$ is decreasing in T ,

$$P_j(X \leq x) = P_j(T \geq \sqrt{1 - z^2}).$$

The three chord-length distribution functions are

$$F_{X,A}(x) = \frac{2}{\pi} \arcsin z, \quad F_{X,B}(x) = 1 - \sqrt{1 - z^2}, \quad F_{X,C}(x) = z^2.$$

They are 0 for $x < 0$ and 1 for $x > 2R$. Substituting $x = \sqrt{3} R$ and taking $1 - F_{X,j}(x)$ gives 1/3, 1/2, 1/4. These continuous laws assign zero probability to the equality at the threshold.

The set-based definitions follow Jagannathan, Chapters 1, 4 and 11. The comparison in common distance and angular coordinates is also developed in *Urban Operations Research*, §3.3.2. The explicit sampling maps and inverse calculations above are provided here in full.

4. The induced law is a probability measure

Let $X: (\Omega, \mathcal{F}, P) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable and define $P_X(B) = P(X^{-1}(B))$. Nonnegativity follows from that of P , and

$$P_X(\mathbb{R}) = P(X^{-1}(\mathbb{R})) = P(\Omega) = 1.$$

If $B_1, B_2, \dots$ are pairwise disjoint Borel sets, their inverse images are pairwise disjoint events, and inverse images preserve their union. Therefore

$$P_X\left(\bigcup_{n=1}^{\infty} B_n\right) = P\left(\bigcup_{n=1}^{\infty} X^{-1}(B_n)\right) = \sum_{n=1}^{\infty} P(X^{-1}(B_n)) = \sum_{n=1}^{\infty} P_X(B_n).$$

This proves that P_X is a probability measure. The same proof applies to the sampling maps Ψ_j and the induced measures P_j on the chord space. Injectivity is unnecessary for this construction: inverse images exist for all functions.

Distinct meanings of the notation f^{-1}

When $f: S \rightarrow T$ is a bijection, $f^{-1}(t)$ denotes an element of S , the value of the inverse function at t . For any subset $B \subseteq T$, the notation $f^{-1}(B)$ denotes a subset of S , the inverse image. The latter is defined even when f is not one-to-one. In the Bertrand construction, C and each Ψ_j are bijections on the stated restricted domains, whereas the chord-length function X is not one-to-one.

5. A subset of the interval to which no probability can be assigned

Section 4 of the main notes states that the event collection for the uniform model on an interval cannot be the set of all subsets. The construction is given here.

The setting

Let $\Omega = [0, 1)$ and define addition modulo 1,

$$x \oplus y = \begin{cases} x + y, & x + y < 1, \\ x + y - 1, & x + y \geq 1, \end{cases}$$

with $\ominus$ the corresponding subtraction. Under $\oplus$ the set $[0, 1)$ is an abelian group; identifying 0 with 1 makes it a circle, and $x \mapsto x \oplus t$ is a rotation. For $A \subseteq [0, 1)$ write $A \oplus t = \{a \oplus t : a \in A\}$.

Suppose P is defined on **every** subset of $[0, 1)$ and satisfies

1. $P([0, 1)) = 1$;
2. countable additivity;
3. rotation invariance: $P(A \oplus t) = P(A)$ for all A and all $t \in [0, 1)$.

The construction

Let $\mathbb{Q}_0 = \mathbb{Q} \cap [0, 1)$, a countable subgroup of $([0, 1), \oplus)$. Define $x \sim y$ when $x \ominus y \in \mathbb{Q}_0$. This is an equivalence relation, and its classes are the cosets $x \oplus \mathbb{Q}_0$.

By the axiom of choice, let $V \subseteq [0, 1)$ contain exactly one element of each class.

Claim 1: the sets $V \oplus q$, for $q \in \mathbb{Q}_0$, are pairwise disjoint. Suppose $v_1 \oplus q_1 = v_2 \oplus q_2$ with $v_1, v_2 \in V$. Then $v_1 \ominus v_2 = q_2 \ominus q_1 \in \mathbb{Q}_0$, so $v_1 \sim v_2$. Since V has one element per class, $v_1 = v_2$, and therefore $q_1 = q_2$.

Claim 2: these sets cover $[0, 1)$. Given $x \in [0, 1)$, let $v \in V$ be the representative of the class of x . Then $q = x \ominus v \in \mathbb{Q}_0$ and $x = v \oplus q \in V \oplus q$.

So $[0, 1)$ is the union of countably many pairwise disjoint sets, each a rotation of V .

The contradiction

Write $c = P(V)$. Rotation invariance gives $P(V \oplus q) = c$ for every q , and countable additivity applied to the countable disjoint family of Claim 1, whose union is $[0, 1)$ by Claim 2, gives

$$1 = P([0, 1)) = \sum_{q \in \mathbb{Q}_0} P(V \oplus q) = \sum_{q \in \mathbb{Q}_0} c.$$

If $c = 0$ the right-hand side is 0; if $c > 0$ it is unbounded. Neither equals 1, so no such P exists, and V is not Lebesgue measurable. This is Vitali's theorem (1905).

Which assumption fails

The argument uses all three conditions, and it is instructive to see which one cannot be weakened.

Countable additivity is the operative assumption in dimension one. Banach (1923) showed that a **finitely additive** rotation-invariant extension of Lebesgue measure to all subsets of the circle does exist.

Restricting to finite additivity therefore removes the obstruction here, at the cost of the limit theorems described in Section 5.5 of the main notes; this is the price of de Finetti's position, noted in Section 5.3 there.

Invariance cannot be dropped without changing the model, since it is what makes the measure uniform. Non-uniform measures on all subsets are unproblematic: a measure concentrated on countably many points is defined on $2^{[0,1]}$ and is countably additive.

Dimension matters. In $\mathbb{R}^3$ even finite additivity is insufficient: the Banach–Tarski theorem (1924) shows that no finitely additive isometry-invariant measure defined on all subsets can assign the unit ball a positive finite value.

Why this does not affect the models used in the course

The selection of V uses the axiom of choice and gives no rule for deciding membership. Solovay (1970) showed it is consistent with Zermelo–Fraenkel set theory together with dependent choice that every set of reals is Lebesgue measurable, so no explicit description of such a set is possible. Every event arising in this lecture — intervals, disks, angular sectors, the preimages $\Psi_j^{-1}(E)$ of Section 3, and countable combinations of these — is a Borel set, and the uniform model is defined on the Borel σ -algebra without difficulty.

6. Exercises on the supplement

1. Verify both $C^{-1}(C(\phi, t)) = (\phi, t)$ and $C(C^{-1}(c)) = c$ on the stated domains. Explain why the restriction on t is needed.
2. Verify both inverse identities for $q_A(u) = \cos(\pi u/2)$ on $(0, 1)$. Repeat for $q_C(u) = \sqrt{u}$.
3. Integrate each density in the comparison table over $(0, 1)$ and over $(0, 1/2)$.
4. Derive the three chord-length distribution functions from the distance distribution functions.
5. Prove that the inverse image of a union is the union of the inverse images. Use this to explain why the induced law is countably additive.
6. For one roll of a die, decide whether each of the following is a sample space, giving a reason: (a) $\{1, \dots, 6\}$; (b) $\{\text{even, odd}\}$; (c) $\{\text{even, greater than 3}\}$; (d) $\{\text{less than 3, equal to 3, greater than 3}\}$.
7. In the Vitali construction, identify the step that fails if $\mathbb{Q}_0$ is replaced by a finite subgroup of $([0, 1), \oplus)$, and state what the argument then yields.
8. A chord generator chooses D/R uniformly on $[0, 1]$. Another chooses a midpoint uniformly by area. Find $P(D < R/3)$ under each.
9. A chord has normalized midpoint distance $T = 1/2$ and midpoint angle $\phi = \pi$. Recover (u, v) under each Ψ_j^{-1} . Does it belong to the strict event $X > \sqrt{3}/2$?
10. In Model B, write the event $T < 1/3$ in terms of $W = T^2$. Calculate its probability using the distribution function of W .

Solutions

1. If $c = C(\phi, t)$, its midpoint is $Rt \sin(\phi)$, so $C^{-1}(c) = (\phi, t)$. Conversely, for $c \in \mathcal{C}_*$, the perpendicular from the circle centre meets the chord at its midpoint and the two endpoints lie a distance $R\sqrt{1-t^2}$ on either side. These are exactly the endpoints in the definition of $C(\phi, t)$, so $C(C^{-1}(c)) = c$. At $t = 0$, the midpoint is zero and its argument is undefined; at $t = 1$, the endpoints coincide. Both cases were excluded and have probability zero under the three laws.
2. On the stated domains, $q_A^{-1}(t) = (2/\pi)\arccost$. Thus $q_A^{-1}(q_A(u)) = u$ and $q_A(q_A^{-1}(t)) = t$. For Model C, $q_C^{-1}(t) = t^2$, so $(\sqrt{u})^2 = u$ and $\sqrt{t^2} = t$ because $t > 0$.

3. For A, an antiderivative is $(2/\pi)\arcsint$, giving total mass 1 and mass $1/3$ below $1/2$. For B, the corresponding values are $[t]_0^1 = 1$ and $[t]_0^{1/2} = 1/2$. For C, they are $[t^2]_0^1 = 1$ and $[t^2]_0^{1/2} = 1/4$.
4. Substitute $a = \sqrt{1 - (x/(2R))^2}$ into $1 - F_{T,j}(a)$ for $0 \leq x \leq 2R$. Use $\arccosa = \arcsin(x/(2R))$ in Model A. The variables have continuous laws, so the endpoint inequalities do not change the probabilities.
5. An outcome belongs to $f^{-1}(\bigcup_n B_n)$ exactly when its image belongs to at least one B_n , equivalently when it belongs to at least one $f^{-1}(B_n)$. The proof in Section 4 then applies.
6. a. Yes: exactly one face shows. (b) Yes: the two are exhaustive and cannot both occur. (c) No: both occur when the face is 4 or 6, so the elements are not mutually exclusive. (d) Yes: the three are exhaustive and pairwise disjoint. For a fair die the two elements of (b) have probability $1/2$ each, whereas those of (d) have probabilities $2/6$, $1/6$ and $3/6$. Both are sample spaces; equal probabilities are not part of the requirement.
7. Claim 1 and Claim 2 still hold, so $[0, 1)$ is a finite disjoint union of n rotations of V and $1 = nc$, giving $c = 1/n$. No contradiction arises: a finite subgroup produces a partition into finitely many sets of equal measure, which is consistent. The contradiction depends on the index set being countably infinite, which is where countable additivity enters.
8. Uniform radial distance gives $1/3$; uniform area gives $(1/3)^2 = 1/9$.
9. $v = 1/2$ in all three models. Model A gives $u = 2/3$, Model B gives $u = 1/2$, and Model C gives $u = 1/4$. The chord has length exactly $\sqrt{3} R$, so it is excluded from the strict event in every case.
10. The event is $W < 1/9$. Its probability is $P(W < 1/9) = \sqrt{1/9} = 1/3$. Assigning probability $1/9$ would incorrectly treat W as uniform under Model B.